

9. (a) A student carries out a series of chemical tests on solutions of three unknown compounds, **A**, **B** and **C**. Her results are recorded in the table.

	A	B	C
Add dilute HCl	no reaction	fizzes	no reaction
Add $\text{BaCl}_2(\text{aq})$	white precipitate forms	no reaction	no reaction
Add $\text{NaOH}(\text{aq})$	green precipitate forms	pungent smelling gas given off, turns damp red litmus paper blue	no reaction
Add $\text{AgNO}_3(\text{aq})$	no reaction	no reaction	cream precipitate forms
Flame test	no colour	no colour	apple-green flame

Use the information provided to give the **chemical name** for each of the compounds. [3]

Compound **A**

Compound **B**

Compound **C**



(b) A technician wants to prepare 250 cm^3 of a 0.25 mol/dm^3 solution of lead nitrate, $\text{Pb}(\text{NO}_3)_2$.

(i) Calculate the number of moles of lead nitrate required to make the solution. [2]

Number of moles = mol

(ii) Calculate the mass of solid lead nitrate that should be dissolved to make the solution. [2]

$A_r(\text{Pb}) = 207$

$A_r(\text{O}) = 16$

$A_r(\text{N}) = 14$

Mass = g

(iii) The only electronic balance available to the technician has a precision of $\pm 0.01\text{ g}$.

Exactly how much lead nitrate should the technician weigh out to ensure that the concentration of the solution is as close as possible to 0.25 mol/dm^3 ? [1]

Mass g

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



Group

→

2

4



9

1

O

[illegible]

Key

relative atomic mass

atomic number